

Briefing for the 4pm Meeting with Shaun

Everything done so far, the numbers to know, and what the meeting will cover

Friday, 31 July 2026

The whole project in one paragraph

We set out to reproduce Zivich & Breskin’s simulation study showing that machine-learning confounding adjustment biases plain g-computation while doubly-robust, cross-fit estimators stay reliable. Exact reproduction needed months of computing, so with Shaun we agreed a lighter design that answers the same question; it reproduced the paper’s message clearly (and its parametric rows match the published numbers almost exactly). Shaun then focused the project on *g-computation’s failure*, and asking *where* the bias lives revealed the project’s best finding: under machine learning, nearly all of it sits in $E(Y^0)$, not $E(Y^1)$. Three further studies isolating why are running now and finish over the weekend.

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1 The numbers to know cold

Truths: $ATE = -0.108$, $E(Y^1) = 0.234$, $E(Y^0) = 0.342$. All results: 1,000 simulated datasets, $n = 3,000$ (except study 2), same datasets throughout, no failed datasets anywhere.

1.1 The headline table (our reproduction of Z&B’s Table 3, ATE)

Coverage should be 95%. “ML” means nuisance models estimated by the super learner.

Estimator	True spec	Main-effects	Machine learning
G-computation	94.8%	76.8%	64.0%
IPW	94.4%	86.7%	93.7%
AIPW	94.1%	85.4%	90.8%
TMLE	93.8%	85.9%	90.1%
SC-AIPW	95.1%	90.4%	93.5%
SC-TMLE	94.6%	92.1%	93.8%

G-computation’s ML bias is $+0.026$ — **matching the paper’s published 0.026 exactly**. Its 64% coverage uses the *empirical* SE, so the failure is pure bias, not bad variance estimation. Cross-fitting also repairs AIPW/TMLE’s slightly-low own-SE coverage ($\sim 90\% \rightarrow \sim 94\%$).

1.2 The arms: where the bias actually lives

G-computation only, machine-learning specification, bias (coverage):

	$E(Y^1)$	$E(Y^0)$	ATE
Baseline (single model, as Z&B)	+0.014 (82.7%)	−0.013 (75.1%)	+0.026 (64.0%)
Study 1: random forest alone	+0.025 (61.0%)	−0.014 (66.9%)	+0.039 (30.0%)
Study 2: $n = 1500$	+0.023 (78.5%)	−0.014 (82.8%)	+0.037 (59.8%)
Study 3: two separate models	−0.001 (95.8%)	−0.017 (68.6%)	+0.017 (85.3%)

The three findings, in speaking order

1. In the single-model version the two arm biases point in opposite directions and ADD in the ATE ($+0.014 - (-0.013) = +0.026$): the ATE hides the structure. **2. Fitting the two outcome models separately (study 3) fixes $E(Y^1)$ completely but leaves $E(Y^0)$ biased** — so the remaining problem is entirely in estimating the untreated mean. This is why Shaun now wants everything focused on $E(Y^0)$. **3. The random forest alone (study 1) is far worse than the super learner** (coverage 30% vs 64%): the super learner’s other members were restraining it. And halving n (study 2) also worsens bias — the “slow convergence” story made visible.

2 What we did, in order

2.1 Phase 1 — exact reproduction attempt

- Forked and ran the authors’ code **byte-for-byte unmodified** (their 9 source files verified pristine with git throughout the entire project).
- Modern libraries break their 2019–20 code in three places (scikit-learn removed `penalty='none'`; statsmodels link-class call; a pandas idiom). We fixed the *environment*, not the code: a pinned Python 3.9 stack on which it runs as-is.
- Their own example script then passed end-to-end (all 11 estimator blocks), and 9 of their 18 production runs completed locally, all matching the paper’s behaviour.
- The remaining runs (super-learner-heavy) needed $\sim 30,000$ core-hours — months on a laptop. The authors themselves used a cluster.

2.2 Phase 2 — the agreed reduced design

Shaun’s redesign, run on 1,000 datasets: keep IPW, AIPW, TMLE as-is; g-computation *without* its 250-fold bootstrap (confidence intervals from the empirical SE instead — the SD of the 1,000 estimates); replace the paper’s unusual “double” cross-fitting with standard single 5-fold cross-fitting, one split. 26 super-learner fits per dataset instead of $\sim 1,456$. Took ~ 4 days on the laptop. Delivered: our version of Z&B’s Table 3 (above) plus histograms of all 1,000 estimates for every estimator (his request, to check normality and outliers — all near-normal, a few isolated outliers in two parametric cells).

2.3 Phase 3 — g-computation focus: baseline arms + studies 1–3

Shaun narrowed the project to g-computation’s failure and asked for $E(Y^1)$ and $E(Y^0)$ separately. We ran the baseline (single-model, as Z&B) recording both arms, plus three one-change-at-a-time studies (random forest only; $n = 1500$; two separate outcome models), each with tables and histograms for all three estimands. Results in §1.2.

2.4 Phase 4 — running now: studies 3c, 4, 5 ($E(Y^0)$ only)

His current request, launched last night and finishing over the weekend:

- **All six estimators, but only for $E(Y^0)$** , always with the arm-specific outcome model: $E(Y | X=0, Z)$ is fitted **using untreated individuals only** (never any $X=1$ data — his key requirement), and $E(Y | X=1, Z)$ is never estimated at all.
- Nuisance models fitted **once per dataset and shared** across estimators: 1 propensity + 1 untreated-outcome model on the full sample, plus one of each per fold for the cross-fit versions — exactly his “12 super learners per dataset”.
- Study 3 continued = $n=3000$ with super learner (~ 1.5 days, running); study 4 = $n=1500$ (~ 1 day); study 5 = random forest (~ 2 hours).
- Because no package computes arm-specific estimates in this form, the estimating equations were written by us — and **validated against the authors’ own code on identical inputs: agreement to machine precision ($\sim 10^{-16}$)** for IPW, AIPW and TMLE before anything was launched.
- Early signal from the first datasets (preliminary!): the doubly-robust estimators appear to repair the $E(Y^0)$ bias that g-computation shows.

3 What the meeting will cover

From his emails, Shaun has explicitly promised to explain two things:

1. **Why he cares about $E(Y^1)$ and $E(Y^0)$ rather than the ATE** (“I’ll explain why when we meet”).
2. **Why the results spared us the algebra** he thought study 4 (old numbering) would need (“The results you have are good enough. I can explain why when we meet.”).

Plus, from earlier in the thread: **the 11 August presentation** (he sent details — re-read that email before the meeting) and **next steps** after studies 3c/4/5.

A plausible explanation to have in mind (verify with him, don’t assert)

Why would the bias sit in $E(Y^0)$? The untreated-outcome model is *fitted* on untreated people but *averaged* over everyone — including the treated, who are systematically sicker (that is the confounding). So estimating $E(Y^0)$ requires the model to predict well in covariate regions where untreated data is thin: an extrapolation problem, which regularised machine learning handles poorly (it shrinks toward the average where data is scarce). $E(Y^1)$ may be easier here because of how the treated/untreated covariate distributions overlap in this particular simulation. If Shaun’s explanation differs, his is the one to take away.

3.1 Worth raising yourself

- The **single cross-fit structure**: Zivich’s implementation trains each nuisance on *one* fold ($n/5$), not the complementary four-fifths as in textbook k -fold cross-fitting. Same fit count, valid, but less training data per model. Our new studies follow the same convention (nuisances from the previous fold). Does he want the textbook version at some point?
- A **brittleness finding**: the authors’ super learner intermittently fails (its weight optimiser errors out; their cross-validation folds are unshuffled). Our runs retry with permuted rows and, if a dataset persists in failing, fall back to logistic regression for that single nuisance fit, loudly logged — a handful of fits out of $\sim 12,000$ at most. Worth one sentence of transparency.
- **Timing honesty**: study 3c is ~ 1.5 days, not the “about a day” first estimated; everything still finishes over the weekend.
- What should the **presentation** emphasise — the reproduction, or the $E(Y^0)$ localisation finding?

4 How things were done (methods you may be asked about)

- **Never touching their code**: all 9 authors’ files pristine (git-verified); our additions are separate driver scripts. Where their scripts needed the `setup` digit changed, we used copies proven identical except that digit.
- **Truths**: ATE -0.1081508 is Z&B’s own hard-coded value (mean of realised $Y^1 - Y^0$ over all 6M rows). Arm truths (0.2339, 0.3419) computed by us as the mean of the DGM’s outcome *probabilities* over the saved confounders; their difference recovers Z&B’s ATE truth to 10^{-4} — a built-in check.
- **Reproducibility**: the data generation is seeded by the authors (datasets identical for everyone); their estimation is *not* seeded (random forest, neural net), so their ML results are not exactly reproducible by anyone — ours are, because our drivers seed per dataset (`seed = 20260721 + sim id`).
- **Confidence intervals**: g-computation has no analytic SE, so its CIs use the empirical SE (Shaun’s convention). Other estimators report their own influence-function SEs; tables show coverage both ways where relevant.
- **Robust running**: results written after every dataset (a crash loses at most one); processes detached from the terminal and protected from system sleep; re-running the same command resumes automatically. (The one overnight loss was the laptop’s battery running out — keep it plugged in.)

- **Where everything lives:** the GitHub fork contains every result CSV, table, histogram PDF, run log, the full decision log (PROJECT_LOG.md), and the 29-page code reference explaining the authors' code file by file.

5 Mini-glossary (30-second refresh)

- **ATE:** average treatment effect, $E(Y^1) - E(Y^0)$; risk difference here.
- $E(Y^0)$: mean outcome if *nobody* were treated — the estimand the project now centres on.
- **g-computation:** fit an outcome model, predict everyone's outcome under a treatment level, average. A plug-in — no bias correction.
- **IPW:** weight people by 1/probability of the treatment they actually got.
- **AIPW / TMLE:** doubly-robust estimators combining both models; correct if either model is right, and they de-bias machine-learning fits.
- **Cross-fitting:** never let a person's estimate use a model trained on them. "SC-" = single cross-fit, 5 folds, one split.
- **Super learner:** cross-validated weighted blend of logistic regression, two GAMs, a 500-tree random forest, a small neural net, and the mean.
- **ASE vs ESE:** the SE the method *claims*, vs the actual SD of the 1,000 estimates. Agreement = honest uncertainty.
- **Coverage:** how often the 95% interval contains the truth; 95% is honest, 64% means misleading conclusions one time in three.

All numbers verified against the raw per-dataset results before this briefing was written. Sources: `table3_reproduction.csv`, `baseline_tables.csv`, `study{1,2,3}_tables.csv`, and the run logs in the repository.